

Project Demonstration Scalability & Future Plan

Date	10 July 2026
Team ID	
Project Name	Smart Lender – AI-Powered Loan Approval Prediction System
Maximum Marks	

Current System Limitations

S.No	Limitation	Impact	Priority
1	Dataset size limited to 614 records	Predictions may lack accuracy for diverse applicant profiles	High
2	KNN accuracy capped at 84.55%	Some borderline applications may be misclassified	High
3	No authentication layer	No role-based access between applicants and loan officers	Medium

Scalability Plan

S.No	Aspect	Current State	Proposed Upgrade
1	Data	Static CSV (loan_data.csv)	Connect to a live database of loan applications for continuous retraining
2	Model	Single deployed KNN model	Periodic re-comparison and re-selection among Decision Tree, Random Forest, KNN, XGBoost
3	Hosting	Local / single-instance Flask app	Deploy to a scalable cloud platform with auto-scaling
4	Security	Basic input validation	Add authentication, HTTPS and role-based access control

Future Roadmap

Phase	Planned Enhancement	Expected Impact
Phase 2	Add explainable-AI (feature-level reasons) to the Result page	Greater transparency for applicants and loan officers
Phase 3	Expand and refresh the training dataset	Improve prediction accuracy beyond 84.55%
Phase 4	Add authentication & loan-officer dashboard	Secure, role-based access and portfolio-level insights